

Virtual element methods for a class of fully nonlinear elliptic PDEs

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1 Introduction

TODO.

2 The continuous problem

Let us denote by $\mathcal{S}_d$ (respectively $\mathcal{S}_d^{++}$) the set of symmetric (respectively symmetric positive definite) matrices of size d . We equip the space $\mathcal{S}_d$ with the Frobenius norm and inner product.

We look for a strong solution $u \in H^2(\Omega)$ to the Isaacs equation

$$\inf_{\alpha \in \mathcal{A}} \sup_{\beta \in \mathcal{B}} [A^{\alpha, \beta}(x) : \nabla^2 u(x) + b^{\alpha, \beta}(x) \cdot \nabla u(x) - c^{\alpha, \beta}(x)u(x) - f^{\alpha, \beta}(x)] = 0 \quad \text{a.e. in } \Omega, \quad (1)$$

where $\Omega \subset \mathbb{R}^d$ is a bounded convex polytope, $\mathcal{A}$ and $\mathcal{B}$ are compact metric spaces, and $A: \mathcal{A} \times \mathcal{B} \times \Omega \rightarrow \mathcal{S}_d^{++}$, $b: \mathcal{A} \times \mathcal{B} \times \Omega \rightarrow \mathbb{R}^d$, $c: \mathcal{A} \times \mathcal{B} \times \Omega \rightarrow \mathbb{R}_+$, and $f: \mathcal{A} \times \mathcal{B} \times \Omega \rightarrow \mathbb{R}$ are functions satisfying the Carathéodory conditions, i.e. they are measurable with respect to $x \in \Omega$ for every $(\alpha, \beta) \in \mathcal{A} \times \mathcal{B}$ and continuous with respect to $(\alpha, \beta) \in \mathcal{A} \times \mathcal{B}$ for almost every $x \in \Omega$. We impose the Dirichlet boundary condition

$$u = g \quad \text{a.e. on } \partial\Omega, \quad (2)$$

for some given $g \in H^2(\Omega)$.

We assume that A , b , and c satisfy the *Cordes condition*:

Assumption 2.1 (Cordes condition). Either one of the two following properties holds:

- There exist $\lambda > 0$ and $\varepsilon \in (0, 1]$ such that for any $\alpha \in \mathcal{A}$, $\beta \in \mathcal{B}$, and $x \in \Omega$, one has

$$\frac{|A^{\alpha, \beta}(x)|^2 + |b^{\alpha, \beta}(x)|^2 / (2\lambda) + (c^{\alpha, \beta}(x)/\lambda)^2}{(\text{tr } A^{\alpha, \beta}(x) + c^{\alpha, \beta}(x)/\lambda)^2} \leq \frac{1}{d + \varepsilon}. \quad (3)$$

- There exists $\varepsilon \in (0, 1]$ such that for any $\alpha \in \mathcal{A}$, $\beta \in \mathcal{B}$, and $x \in \Omega$, one has

$$\frac{|A^{\alpha, \beta}(x)|^2}{(\text{tr } A^{\alpha, \beta}(x))^2} \leq \frac{1}{d - 1 + \varepsilon}, \quad b^{\alpha, \beta}(x) = 0, \quad c^{\alpha, \beta}(x) = 0. \quad (4)$$

In this case, we let $\lambda := 0$.

TODO:
check this
is the right
assump-
tion.

23 We will not use the Cordes condition directly, but rather its following consequence.

24 **Proposition 2.2.** *There exist $\lambda \geq 0$, $\kappa \in (0, 1]$, and a function $\gamma: \mathcal{A} \times \mathcal{B} \times \Omega \rightarrow \mathbb{R}_+^*$, measurable*
 25 *with respect to $x \in \Omega$ for every $(\alpha, \beta) \in \mathcal{A} \times \mathcal{B}$ and continuous with respect to $(\alpha, \beta) \in \mathcal{A} \times \mathcal{B}$*
 26 *for almost every $x \in \Omega$, such that, for any $\alpha \in \mathcal{A}$, $\beta \in \mathcal{B}$, and $x \in \Omega$,*

$$\lambda^2 |\gamma^{\alpha, \beta}(x) A^{\alpha, \beta}(x) - I_d|^2 + \lambda |\gamma^{\alpha, \beta}(x) b^{\alpha, \beta}(x)|^2 / 2 + |\gamma^{\alpha, \beta}(x) c^{\alpha, \beta}(x) - \lambda|^2 \leq \lambda^2 (1 - \kappa)^2, \quad (5)$$

27 and in the case $\lambda = 0$,

$$|\gamma^{\alpha, \beta}(x) A^{\alpha, \beta}(x) - I_d|^2 \leq (1 - \kappa)^2, \quad b^{\alpha, \beta}(x) = 0, \quad c^{\alpha, \beta}(x) = 0. \quad (6)$$

28 *Proof.* It is easily verified that when letting $\kappa := 1 - \sqrt{1 - \varepsilon}$ and

$$\gamma^{\alpha, \beta}(x) := \begin{cases} \left(\operatorname{tr} A^{\alpha, \beta}(x) + \frac{c^{\alpha, \beta}(x)}{\lambda} \right) / \left(|A^{\alpha, \beta}(x)|^2 + \frac{|b^{\alpha, \beta}(x)|^2}{2\lambda} + \frac{c^{\alpha, \beta}(x)^2}{\lambda^2} \right) & \text{if } \lambda > 0, \\ \operatorname{tr} A^{\alpha, \beta}(x) / |A^{\alpha, \beta}(x)|^2 & \text{if } \lambda = 0, \end{cases} \quad (7)$$

29 the inequalities (5) and (6) reduce respectively to (3) and (4). $\square$

30 *Remark 2.3.* For any $\alpha \in \mathcal{A}$, $\beta \in \mathcal{B}$, and $x \in \Omega$, the quantity $\gamma^{\alpha, \beta}(x)$ defined in (7) is minimizing
 31 the left-hand side of (6) if $\lambda > 0$, and is minimizing the left-hand side of the first inequality in (5)
 32 if $\lambda = 0$, as can be verified by writing the relevant first-order optimality condition. Therefore,
 33 the statement of Proposition 2.2 is actually a characterization of the Cordes condition. The
 34 presence of the additional conditions (6) in the case $\lambda = 0$ can be interpreted as follows: if $\lambda > 0$,
 35 then (5) implies that $|\gamma^{\alpha, \beta}(x) A^{\alpha, \beta}(x) - I_d|^2 \leq (1 - \kappa)^2$, $|\gamma^{\alpha, \beta}(x) b^{\alpha, \beta}(x)|^2 / 2 \leq \lambda(1 - \kappa)^2$, and
 36 $|\gamma^{\alpha, \beta}(x) c^{\alpha, \beta}(x) - \lambda|^2 \leq \lambda^2(1 - \kappa)^2$; (6) results from letting λ approach zero in these inequalities.

37 From now on, we let $\lambda \geq 0$, $\kappa \in (0, 1]$, and $\gamma: \mathcal{A} \times \mathcal{B} \times \Omega \rightarrow \mathbb{R}$ be as in Proposition 2.2. In
 38 practice, the function γ may be chosen according to (7), although we also allow other choices of
 39 functions γ as long as the inequalities (5) and (6) are satisfied.

40 One consequence of (5) and (6) is that the functions

$$x \mapsto \sup_{\substack{\alpha \in \mathcal{A} \\ \beta \in \mathcal{B}}} \gamma^{\alpha, \beta}(x) |A^{\alpha, \beta}(x)|, \quad x \mapsto \sup_{\substack{\alpha \in \mathcal{A} \\ \beta \in \mathcal{B}}} \gamma^{\alpha, \beta}(x) |b^{\alpha, \beta}(x)|, \quad x \mapsto \sup_{\substack{\alpha \in \mathcal{A} \\ \beta \in \mathcal{B}}} \gamma^{\alpha, \beta}(x) c^{\alpha, \beta}(x),$$

41 all belong to $L^\infty(\Omega)$. We need to assume a similar condition on the source term f :

42 **Assumption 2.4** (source term). The function $x \mapsto \sup_{\alpha \in \mathcal{A}, \beta \in \mathcal{B}} \gamma^{\alpha, \beta}(x) |f^{\alpha, \beta}(x)|$ belongs to
 43 $L^2(\Omega)$.

44 In this setting, the analysis of the Dirichlet problem can be performed by relying on an
 45 equivalent variational formulation. In order to describe it, let us first define, for any $v \in H^2(\Omega)$,
 46 the pointwise operators

$$\begin{aligned} \mathcal{L}^{\alpha, \beta} v &:= A^{\alpha, \beta} : \nabla^2 v + b^{\alpha, \beta} \cdot \nabla v - c^{\alpha, \beta} v, & \mathcal{L}_\lambda v &:= \Delta v - \lambda v, \\ F[v] &:= \inf_{\alpha \in \mathcal{A}} \sup_{\beta \in \mathcal{B}} [\mathcal{L}^{\alpha, \beta} v - f^{\alpha, \beta}], & F_\gamma[v] &:= \inf_{\alpha \in \mathcal{A}} \sup_{\beta \in \mathcal{B}} [\gamma^{\alpha, \beta} (\mathcal{L}^{\alpha, \beta} v - f^{\alpha, \beta})] \end{aligned} \quad (8)$$

47 (in our analysis, we will also apply these operators to functions belonging to some broken H^2
 48 space; we naturally extend their definitions to this case). Then the Isaacs equation (1) can be
 49 written as $F[u] = 0$ a.e. in Ω , and is equivalent to $F_\gamma[u] = 0$ a.e. in Ω . Moreover, it is easily
 50 verified that $F_\gamma[v] \in L^2(\Omega)$ for any $v \in H^2(\Omega)$.

Let us then introduce the operator $\mathcal{A}: H^2(\Omega) \rightarrow (H^2(\Omega))^*$ defined by

$$\langle \mathcal{A}(v), w \rangle := \int_{\Omega} F_{\gamma}[v] \mathcal{L}_{\lambda} w \, dx, \quad (9)$$

and the spaces

$$V^g := \{v \in H^2(\Omega) \mid v|_{\partial\Omega} = g|_{\partial\Omega}\}, \quad V^0 := H^2(\Omega) \cap H_0^1(\Omega) = \{v \in H^2(\Omega) \mid v|_{\partial\Omega} = 0\}.$$

Then the problem (1), with the boundary condition (2), is equivalent to the variational problem

$$\text{find } u \in V^g \text{ s.t. } \forall v \in V^0, \langle \mathcal{A}(u), v \rangle = 0. \quad (10)$$

The equivalence follows from the fact that $\{\mathcal{L}_{\lambda} v \mid v \in V^0\} = L^2(\Omega)$, so testing against $\mathcal{L}_{\lambda} v$ for any $v \in V^0$ amounts to testing against any L^2 function.

Under our assumptions, it is well-known that there exists a unique solution $u \in V^g$ to (10). One may refer to [2] for a proof, or, equivalently, follow the sketch of the proof that we use in section 5 for the well-posedness of our discrete variational problem (15).

Technically more regularity of the coefficients is assumed in [2], but I don't think this regularity is used in the proof.

3 Virtual element framework

In this section, we describe the H^2 conforming virtual element framework that we will use in section 4 as a basis for our discretization of the variational problem (10).

We assume that we are given a polytopal mesh $\mathcal{K}_h$ discretizing Ω , and an arbitrary polynomial consistency order $m \geq 2$. We associate some characteristic length $h_F > 0$ to each d' -dimensional face of each cell $K \in \mathcal{K}_h$, for $0 \leq d' \leq d$, and assume that the following standard shape-regularity assumption holds.

Assumption 3.1 (shape-regularity). There exists $\rho \in (0, 1)$ such that any d' -dimensional face F of any $K \in \mathcal{K}_h$, with $0 \leq d' \leq d$, is star-shaped with respect to a d' -dimensional ball of diameter $\rho \text{diam}(K)$, and its characteristic length satisfies $\rho \text{diam}(K) \leq h_F \leq \rho^{-1} \text{diam}(K)$.

From now on, for any two quantities $a_1, a_2 \in \mathbb{R}$, we write $a_1 \lesssim a_2$ (respectively $a_1 \gtrsim a_2$) if there exists some constant $C > 0$, depending only on d, Ω , the polynomial consistency order m , and the shape-regularity parameter ρ , such that $a_1 \leq C a_2$ (respectively $a_1 \geq C a_2$). We write $a_1 \simeq a_2$ if both $a_1 \lesssim a_2$ and $a_1 \gtrsim a_2$.

We let $\Omega_h := \bigcup_{K \in \mathcal{K}_h} \text{int}(K)$, in order to be able to define the broken H^2 space $H^2(\Omega_h)$; observe that $H^2(\Omega) \subset H^2(\Omega_h) \subset L^2(\Omega)$. In our analysis, we will rely on the seminorms $|\cdot|_{*,\lambda,h}$ and $|\cdot|_{*,\lambda}$ defined respectively on $H^2(\Omega_h)$ and $H^2(\Omega)$ by

$$|v|_{*,\lambda,h}^2 := |v|_{2,h}^2 + 2\lambda |v|_{1,h}^2 + \lambda^2 \|v\|_{0,h}^2, \quad |v|_{*,\lambda}^2 := |v|_2^2 + 2\lambda |v|_1^2 + \lambda^2 \|v\|_0^2.$$

For any polytope ω (be it of dimension d or less) and any $k \in \mathbb{N}$, we denote by Π_{ω}^k the L^2 projection operator onto the space $\mathbb{P}_k(\omega)$ of polynomials of degree k on ω . For any $k \in \mathbb{N}$, we define the space

$$\mathbb{P}_{k,h}(\Omega) := \{p \in L^2(\Omega) \mid \forall K \in \mathcal{K}_h, p|_K \in \mathbb{P}_k(K)\}$$

of discontinuous piecewise polynomial functions of degree k on the mesh $\mathcal{K}_h$, and denote by Π_h^k the L^2 projection operator onto $\mathbb{P}_{k,h}(\Omega)$. We naturally extend the definitions of the projection operators Π_{ω}^k and Π_h^k to the cases of vector- and matrix-valued functions.

We assume that we are given:

- A finite-dimensional linear subspace $V_h \subset H^2(\Omega)$; we then let $V_h^0 := V_h \cap H_0^1(\Omega)$.
- A function $g_I \in V_h$, to be interpreted as some interpolant of the function g from the Dirichlet boundary condition (2); we then let $V_h^g := g_I + V_h^0$.
- A symmetric bilinear form $s_{*,\lambda,h}: (V_h + \mathbb{P}_{m,h}(\Omega))^2 \rightarrow \mathbb{R}$.

In order to accommodate for the several variants of the H^2 conforming virtual element method described in the literature and implemented in computer codes, we do not prescribe a specific construction of V_h , g_I , and $s_{*,\lambda,h}$, but only impose that the following assumptions hold.

TODO: maybe also refer to another section of the manuscript for an example of a construction.

Assumption 3.2 (inverse inequalities). For any $v \in V_h$, cell $K \in \mathcal{K}_h$, and side S of K ,

$$|v|_{2,K} \lesssim h_K^{-1} |v|_{1,K}, \quad (11)$$

$$|v|_{3/2,S} \lesssim h_S^{-3/2} \|v\|_{0,S}. \quad (12)$$

Assumption 3.3 (stabilization form scaling). For any $v \in V_h$, one has

$$s_{*,\lambda,h}(v - \Pi_h^m v, v - \Pi_h^m v) \approx |v - \Pi_h^m v|_{*,\lambda,h}^2. \quad (13)$$

Remark 3.4 (requirement for inverse inequalities). We will only need the inverse inequalities (11) and (12) in section 6, and not in the analysis of our main numerical scheme (15).

Remark 3.5 (dependency in λ). The construction of the bilinear form $s_{*,\lambda,h}$ is expected to depend on λ . For instance, one may choose to define $s_{*,\lambda,h}(v, w)$ as $s_{2,h}(v, w) + 2\lambda s_{1,h}(v, w) + \lambda^2 s_{0,h}(v, w)$, where the $s_{i,h}$, $0 \leq i \leq 2$, are symmetric bilinear forms satisfying $s_{i,h}(v - \Pi_h^m v, v - \Pi_h^m v) \approx |v - \Pi_h^m v|_{i,h}^2$ for $v \in V_h$, that may be built in a standard way.

One important feature of the virtual element method is that functions of V_h are typically *virtual*, in the sense that their pointwise values cannot be efficiently computed from the values of the degrees of freedom that characterize them. Rather, we only assume the efficient computability of the discontinuous piecewise polynomial functions $\Pi_h^m v$, $\Pi_h^m \nabla v$, and $\Pi_h^m \nabla^2 v$, and of $s_{*,\lambda,h}(v - \Pi_h^m v, w - \Pi_h^m w)$, for $v, w \in V_h$.

Remark 3.6 (degrees of the projections). Choosing the degree m for all projections is nonstandard: usually, for H^2 conforming virtual element functions, only the efficient computability of $\Pi_h^{m-1} \nabla v$ and $\Pi_h^{m-2} \nabla^2 v$ is assumed, but this is not enough in our setting. We claim that assuming the efficient computability of $\Pi_h^m \nabla v$ and $\Pi_h^m \nabla^2 v$ is not very restrictive. Following standard virtual element principles, we expect these projections to be computed from integration by parts formulas, which can be applied provided that $\Pi_K^m v$, $\Pi_S^m v$, $\Pi_S^m \partial_{\mathbf{n}} v$, and $\Pi_R^m v$ are efficiently computable on any cell $K \in \mathcal{K}_h$, side S of K , and side R of S (in the terminology of polytopes, S and R are respectively a *facet* and a *ridge* of K). The only difference with the standard setting concerns the projection $\Pi_S^m \partial_{\mathbf{n}} v$: in dimension $d > 2$, standard H^2 conforming virtual spaces are designed so that $\Pi_S^{m-1} \partial_{\mathbf{n}} v$ is efficiently computable, which is missing an order. However, following [1, Remark 5], one may easily adapt the construction of V_h so that $\Pi_S^m \partial_{\mathbf{n}} v$ is also efficiently computable, without any change to the degrees of freedom associated to V_h . In dimension two, S is an edge and $\partial_{\mathbf{n}} v$ is typically polynomial on S , in which case this adaptation is not even required.

4 The virtual element scheme

Since the quantity $\langle \mathcal{A}(v), w \rangle$ defined in (9) is typically not efficiently computable from the values of the degrees of freedom associated to functions $v \in V_h^g$ and $w \in V_h^0$, we need to design a variant of the operator $\mathcal{A}$ suitable for use in our virtual element scheme.

First, we define, for $v \in H^2(\Omega_h)$, the following variants of the operators $\mathcal{L}^{\alpha,\beta}$, $\mathcal{L}_\lambda$, and F_γ defined in (8):

$$\begin{aligned}\mathcal{L}_h^{\alpha,\beta} v &:= A^{\alpha,\beta} : \Pi_h^m \nabla^2 v + b^{\alpha,\beta} \cdot \Pi_h^m \nabla v - c^{\alpha,\beta} \Pi_h^m v, \\ \mathcal{L}_{\lambda,h} v &:= \Pi_h^m (\Delta v - \lambda v), \\ F_{\gamma,h}[v] &:= \inf_{\alpha \in \mathcal{A}} \sup_{\beta \in \mathcal{B}} [\gamma^{\alpha,\beta} (\mathcal{L}_h^{\alpha,\beta} v - f^{\alpha,\beta})],\end{aligned}\tag{14}$$

where ∇ , ∇^2 , and Δ denote respectively the broken gradient, Hessian, and Laplacian.

We then define operators $\mathcal{A}_h^c$, $\mathcal{A}_h^s$, and $\mathcal{A}_h$ from $V_h + \mathbb{P}_{m,h}(\Omega)$ to $(V_h)^*$ by letting, for $v \in V_h + \mathbb{P}_{m,h}(\Omega)$ and $w \in V_h$,

$$\langle \mathcal{A}_h^c(v), w \rangle := \int_{\Omega} F_{\gamma,h}[v] \mathcal{L}_{\lambda,h} w \, dx, \quad \langle \mathcal{A}_h^s(v), w \rangle := s_{*,\lambda,h}(v - \Pi_h^m v, w - \Pi_h^m w),$$

and

$$\langle \mathcal{A}_h(v), w \rangle := \langle \mathcal{A}_h^c(v), w \rangle + \eta \langle \mathcal{A}_h^s(v), w \rangle,$$

where $\eta > 0$ is some given stabilization factor.

Remark 4.1. In the definition of our numerical scheme, we only need to apply the operators $\mathcal{A}_h^c$, $\mathcal{A}_h^s$, and $\mathcal{A}_h$ to functions of V_h , but it will be useful in our analysis that they are also defined for functions of $V_h + \mathbb{P}_{m,h}(\Omega)$.

We consider the virtual element scheme:

$$\text{find } u_h \in V_h^g \text{ s.t. } \forall v \in V_h^0, \langle \mathcal{A}_h(u_h), v \rangle = 0.\tag{15}$$

5 Analysis of the scheme

In this section, we prove that the existence of a unique solution to the scheme (15) (Theorem 5.6) and an error estimate for this solution (Theorem 5.7), provided that the stabilization factor η is large enough. Let us make this assumption precise through the following definition:

Definition 5.1. We say that a proposition (P) holds for η large enough if there exists a constant $C \lesssim 1$ such that (P) holds whenever $\eta \geq C$.

Following the literature on nondivergence form elliptic equations with Cordes coefficients, we interpret the nonlinear nondivergence form operator $F_{\gamma,h}$ as a perturbation of the linear, divergence form operator $\mathcal{L}_{\lambda,h}$. This introduces an error that can be controlled as follows.

Lemma 5.2 (linear approximation). *For any $v_1, v_2 \in H^2(\Omega)$,*

$$\|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h}(v_1 - v_2)\|_0 \leq (1 - \kappa) |v_1 - v_2|_{*,\lambda}.\tag{16}$$

142 *Proof.* Let $w := v_1 - v_2$. One has

$$\begin{aligned}
\|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h}(v_1 - v_2)\|_0 &= \left\| \inf_{\alpha_1 \in \mathcal{A}} \sup_{\beta_1 \in \mathcal{B}} [\gamma^{\alpha_1, \beta_1}(\mathcal{L}_h^{\alpha_1, \beta_1} v_1 - f^{\alpha_1, \beta_1}) - \mathcal{L}_{\lambda,h} v_1] \right. \\
&\quad \left. - \inf_{\alpha_2 \in \mathcal{A}} \sup_{\beta_2 \in \mathcal{B}} [\gamma^{\alpha_2, \beta_2}(\mathcal{L}_h^{\alpha_2, \beta_2} v_2 - f^{\alpha_2, \beta_2}) - \mathcal{L}_{\lambda,h} v_2] \right\|_0 \\
&\leq \left\| \sup_{\substack{\alpha \in \mathcal{A} \\ \beta \in \mathcal{B}}} |\gamma^{\alpha, \beta} \mathcal{L}_h^{\alpha, \beta} w - \mathcal{L}_{\lambda,h} w| \right\|_0 \\
&= \left\| \sup_{\substack{\alpha \in \mathcal{A} \\ \beta \in \mathcal{B}}} |(\gamma^{\alpha, \beta} A^{\alpha, \beta} - I_d) : \Pi_h^m \nabla^2 w + \gamma^{\alpha, \beta} b^{\alpha, \beta} \cdot \Pi_h^m \nabla w \right. \\
&\quad \left. - (\gamma^{\alpha, \beta} c^{\alpha, \beta} - \lambda) \Pi_h^m w \right\|_0.
\end{aligned}$$

143 If $\lambda = 0$, we conclude using the Cordes properties (6):

$$\|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h}(v_1 - v_2)\|_0 \leq (1 - \kappa) \|\Pi_h^m \nabla^2 w\|_0 \leq (1 - \kappa) \|\nabla^2 w\|_0 = (1 - \kappa) |w|_{*, \lambda}.$$

144 If $\lambda > 0$, we use the Cauchy-Schwarz inequality and then the Cordes inequality (5):

$$\begin{aligned}
&\|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h}(v_1 - v_2)\|_0 \\
&\leq \left\| \sup_{\substack{\alpha \in \mathcal{A} \\ \beta \in \mathcal{B}}} \left(|\gamma^{\alpha, \beta} A^{\alpha, \beta} - I_d|^2 + \frac{1}{2\lambda} |\gamma^{\alpha, \beta} b^{\alpha, \beta}|^2 + \frac{1}{\lambda^2} |\gamma^{\alpha, \beta} c^{\alpha, \beta} - \lambda|^2 \right)^{1/2} \right. \\
&\quad \left. (|\Pi_h^m \nabla^2 w|^2 + 2\lambda |\Pi_h^m \nabla w|^2 + \lambda^2 |\Pi_h^m w|^2)^{1/2} \right\|_0 \\
&\leq (1 - \kappa) (\|\Pi_h^m \nabla^2 w\|_0^2 + 2\lambda \|\Pi_h^m \nabla w\|_0^2 + \lambda^2 \|\Pi_h^m w\|_0^2)^{1/2} \leq (1 - \kappa) |w|_{*, \lambda},
\end{aligned}$$

145 which concludes the proof. $\square$

146 By similar computations, one can prove the Lipschitz continuity of the operators F_γ and $F_{\gamma,h}$.

147 **Lemma 5.3** (Lipschitz continuity). *For any $v_1, v_2 \in H^2(\Omega_h)$, one has*

$$\|F_\gamma[v_1] - F_\gamma[v_2]\|_0 \lesssim |v_1 - v_2|_{*, \lambda, h}, \quad \|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2]\|_0 \lesssim |v_1 - v_2|_{*, \lambda, h}. \quad (17)$$

148 *Proof.* This is easily verified, using the Cordes inequality (5), or alternatively using (6) in the
149 case $\lambda = 0$, in order to estimate $\gamma^{\alpha, \beta}(x) |A^{\alpha, \beta}(x)|$, $\gamma^{\alpha, \beta}(x) |b^{\alpha, \beta}(x)|$, and $\gamma^{\alpha, \beta}(x) c^{\alpha, \beta}(x)$ for every
150 $\alpha \in \mathcal{A}$, $\beta \in \mathcal{B}$, and $x \in \Omega$. $\square$

151 We also need the following inequality for functions of the space $V^0 = H^2(\Omega) \cap H_0^1(\Omega)$, which
152 we refer to as a Miranda-Talenti inequality since it follows from the standard *Miranda-Talenti*
153 *estimate*, and reduces to it in the case $\lambda = 0$.

154 **Proposition 5.4** (Miranda-Talenti inequality). *For any $v \in V^0$,*

$$|v|_{*, \lambda} \leq \|\mathcal{L}_\lambda v\|_0. \quad (18)$$

155 *Proof.* One has

$$\|\mathcal{L}_\lambda v\|_0^2 = \int_\Omega (\Delta v - \lambda v)^2 dx = \|\Delta v\|_0^2 - 2\lambda \int_\Omega (\Delta v) v dx + \lambda^2 \|v\|_0^2.$$

156 Integrating by parts, $\|\mathcal{L}_\lambda v\|_0^2 = \|\Delta v\|_0^2 + 2\lambda |v|_1^2 + \lambda^2 \|v\|_0^2$. Since Ω is convex and $v \in V^0$, the
157 Miranda-Talenti estimate $|v|_2 \leq \|\Delta v\|_0$ holds (see [3, 4]), which concludes the proof. $\square$

From the linear approximation inequality (16) and the Miranda-Talenti inequality (18), we deduce a strong monotonicity estimate for the operator $\mathcal{A}_h$, and then the existence of a unique solution to the scheme (15).

Lemma 5.5 (strong monotonicity). *For η large enough in the sense of Definition 5.1, it holds for any $v_1, v_2 \in V_h^g$ that*

$$\langle \mathcal{A}_h(v_1) - \mathcal{A}_h(v_2), v_1 - v_2 \rangle \geq \kappa |v_1 - v_2|_{*,\lambda}^2. \quad (19)$$

Proof. Let $w := v_1 - v_2$. One has

$$\begin{aligned} \langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle &= \int_{\Omega} (F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2]) \mathcal{L}_{\lambda,h} w \, dx \\ &= \|\mathcal{L}_{\lambda,h} w\|_0^2 + \int_{\Omega} (F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h} w) \mathcal{L}_{\lambda,h} w \, dx \\ &\geq \|\mathcal{L}_{\lambda,h} w\|_0^2 - \|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h} w\|_0 \|\mathcal{L}_{\lambda,h} w\|_0. \end{aligned}$$

Using (16) to estimate $\|F_{\gamma,h}[v_1] - F_{\gamma,h}[v_2] - \mathcal{L}_{\lambda,h} w\|_0$, and using the fact that $\|\mathcal{L}_{\lambda,h} w\|_0 = \|\Pi_h^m \mathcal{L}_{\lambda} w\|_0 \leq \|\mathcal{L}_{\lambda} w\|_0$,

$$\langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle \geq \|\Pi_h^m \mathcal{L}_{\lambda} w\|_0^2 - (1 - \kappa) |w|_{*,\lambda} \|\mathcal{L}_{\lambda} w\|_0.$$

On the other hand, using the stabilization scaling inequality (13),

$$\langle \mathcal{A}_h^s(v_1) - \mathcal{A}_h^s(v_2), w \rangle \gtrsim |w - \Pi_h^m w|_{*,\lambda,h}^2 \gtrsim \|\mathcal{L}_{\lambda}(w - \Pi_h^m w)\|_0^2 = \|\mathcal{L}_{\lambda} w - \mathcal{L}_{\lambda} \Pi_h^m w\|_0^2,$$

Since $\mathcal{L}_{\lambda} \Pi_h^m w \in \mathbb{P}_{m,h}(\Omega)$, the definition of the L^2 projection implies that $\|\mathcal{L}_{\lambda} w - \Pi_h^m \mathcal{L}_{\lambda} w\|_0^2 \leq \|\mathcal{L}_{\lambda} w - \mathcal{L}_{\lambda} \Pi_h^m w\|_0^2$, hence

$$\langle \mathcal{A}_h^s(v_1) - \mathcal{A}_h^s(v_2), w \rangle \gtrsim \|\mathcal{L}_{\lambda} w - \Pi_h^m \mathcal{L}_{\lambda} w\|_0^2.$$

We can assume η to be large enough so that $\eta \langle \mathcal{A}_h^s(v_1) - \mathcal{A}_h^s(v_2), w \rangle \geq \|\mathcal{L}_{\lambda} w - \Pi_h^m \mathcal{L}_{\lambda} w\|_0^2$, and thus

$$\begin{aligned} \langle \mathcal{A}_h(v_1) - \mathcal{A}_h(v_2), w \rangle &= \langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle + \eta \langle \mathcal{A}_h^s(v_1) - \mathcal{A}_h^s(v_2), w \rangle \\ &\geq \|\Pi_h^m \mathcal{L}_{\lambda} w\|_0^2 - (1 - \kappa) |w|_{*,\lambda} \|\mathcal{L}_{\lambda} w\|_0 + \|\mathcal{L}_{\lambda} w - \Pi_h^m \mathcal{L}_{\lambda} w\|_0^2 \\ &= \|\mathcal{L}_{\lambda} w\|_0^2 - (1 - \kappa) |w|_{*,\lambda} \|\mathcal{L}_{\lambda} w\|_0, \end{aligned}$$

where the last equality follows from the Pythagorean theorem. Using two times successively the Miranda-Talenti inequality (18), $\|\mathcal{L}_{\lambda} w\|_0^2 - (1 - \kappa) |w|_{*,\lambda} \|\mathcal{L}_{\lambda} w\|_0 \geq \kappa \|\mathcal{L}_{\lambda} w\|_0^2 \geq \kappa |w|_{*,\lambda}^2$, which concludes the proof. $\square$

Theorem 5.6 (existence and uniqueness). *For η large enough in the sense of Definition 5.1, there exists a unique solution $u_h \in V_h^g$ to (15).*

Proof. The inequality (19) implies that $\mathcal{A}_h$ is strongly monotone when seen as an operator from V_h^g to $(V_h^0)^*$. It is easily verified that it is also Lipschitz continuous, using the Lipschitz continuity estimate (17) for $F_{\gamma,h}$ and the stabilization scaling inequality (13). Therefore, the Browder-Minty theorem applies, and yields the announced result. $\square$

We end this section by providing the following estimate of the error between the solution $u_h \in V_h^g$ to the scheme (15) and an arbitrary function $u_I \in V_h^g$, which should be understood as some interpolant of the solution $u \in V^g$ to (1) and (2).

Theorem 5.7 (error estimate). *For η large enough in the sense of Definition 5.1, and for any $u_I \in V_h^g$,*

$$|u_I - u_h|_{*,\lambda} \lesssim \eta \kappa^{-1} (|u - u_I|_{*,\lambda} + |u - \Pi_h^m u|_{*,\lambda,h}).$$

Proof. Let $w := u_I - u_h$. Using the strong monotonicity inequality (19), and then the fact that u_h is solution to the scheme (15),

$$|w|_{*,\lambda}^2 \leq \kappa^{-1} \langle \mathcal{A}_h(u_I) - \mathcal{A}_h(u_h), w \rangle = \kappa^{-1} \langle \mathcal{A}_h(u_I), w \rangle.$$

It remains to prove that $\langle \mathcal{A}_h(u_I), w \rangle \lesssim \eta (|u - u_I|_{*,\lambda} + |u - \Pi_h^m u|_{*,\lambda,h}) |w|_{*,\lambda}$.

One has $\langle \mathcal{A}_h(u_I), w \rangle = \langle \mathcal{A}_h(u_I) - \mathcal{A}_h(\Pi_h^m u), w \rangle + \langle \mathcal{A}_h(\Pi_h^m u), w \rangle$. Using the Lipschitz continuity estimate (17), the stabilization scaling inequality (13), and the triangle inequality $|u_I - \Pi_h^m u|_{*,\lambda,h} \leq |u - u_I|_{*,\lambda} + |u - \Pi_h^m u|_{*,\lambda,h}$, it is easy to prove that

$$\langle \mathcal{A}_h(u_I) - \mathcal{A}_h(\Pi_h^m u), w \rangle \lesssim \eta (|u - u_I|_{*,\lambda} + |u - \Pi_h^m u|_{*,\lambda,h}) |w|_{*,\lambda}.$$

It remains to prove that $\langle \mathcal{A}_h(\Pi_h^m u), w \rangle \lesssim (\eta_s + \eta_b) (|u - u_I|_{*,\lambda} + |u - \Pi_h^m u|_{*,\lambda,h}) |w|_{*,\lambda}$.

Since $\Pi_h^m u \in \mathbb{P}_{m,h}(\Omega)$, one has $\langle \mathcal{A}_h^s(\Pi_h^m u), w \rangle = 0$ and $F_{\gamma,h}[\Pi_h^m u] = F_\gamma[\Pi_h^m u]$ almost everywhere. Therefore,

$$\langle \mathcal{A}_h(\Pi_h^m u), w \rangle = \langle \mathcal{A}_h^c(\Pi_h^m u), w \rangle = \int_{\Omega} F_{\gamma,h}[\Pi_h^m u] \mathcal{L}_{\lambda,h} w \, dx = \int_{\Omega} F_\gamma[\Pi_h^m u] \mathcal{L}_{\lambda,h} w \, dx.$$

Since u is a strong solution to (1), one has $F_\gamma[u] = 0$ almost everywhere, so

$$\langle \mathcal{A}_h(\Pi_h^m u), w \rangle = \int_{\Omega} (F_\gamma[\Pi_h^m u] - F_\gamma[u]) \mathcal{L}_{\lambda,h} w \, dx.$$

Using the Lipschitz continuity estimate (17) for F_γ , we deduce that $\langle \mathcal{A}_h(\Pi_h^m u), w \rangle \lesssim |u - \Pi_h^m u|_{*,\lambda,h} |w|_{*,\lambda}$, which concludes the proof. $\square$

6 Variant of the scheme with weak imposition of the boundary condition

In this section, we study a variant of the scheme (15) in which we impose the Dirichlet boundary condition (2) weakly, through a Nitsche-type boundary penalty method.

Let us define the set

$$\mathcal{S}_{\partial,h} := \{S \text{ side of } K \mid K \in \mathcal{K}_h\}$$

of boundary sides of the mesh $\mathcal{K}_h$, the space $\mathbb{P}_{m,h}(\partial\Omega)$ of functions $p \in L^2(\partial\Omega)$ satisfying $p|_S \in \mathbb{P}_m(S)$ on each $S \in \mathcal{S}_{\partial,h}$, the L^2 projection operator $\Pi_{\partial,h}^m$ from $L^2(\partial\Omega)$ to $\mathbb{P}_{m,h}(\partial\Omega)$, and the space $V_{\partial,h} := \{v|_{\partial\Omega} \mid v \in V_h\}$. For any $v \in L^2(\partial\Omega)$, let

$$|v|_{\partial,\lambda,h}^2 := \sum_{S \in \mathcal{S}_{\partial,h}} (h_S^{-3} + \lambda h_S^{-1}) \|v\|_{0,S}^2.$$

We assume that we are given a symmetric bilinear form $s_{\partial,\lambda,h}: (V_{\partial,h} + \mathbb{P}_{m,h}(\partial\Omega))^2 \rightarrow \mathbb{R}$ satisfying the following assumption.

Assumption 6.1 (boundary stabilization form scaling). For any $v \in V_{\partial,h}$, one has

$$s_{\partial,\lambda,h}(v - \Pi_{\partial,h}^m v, v - \Pi_{\partial,h}^m v) \approx |v - \Pi_{\partial,h}^m v|_{\partial,\lambda,h}^2. \quad (20)$$

Let us define operators $\mathcal{A}_h^b$ and $\mathcal{A}_h^w$ from $V_h + \mathbb{P}_{m,h}(\Omega)$ to $(V_h)^*$ by

$$\begin{aligned}\langle \mathcal{A}_h^b(v), w \rangle &:= \sum_{S \in \mathcal{S}_{\partial,h}} (h_S^{-3} + \lambda h_S^{-1}) \int_S (\Pi_S^m v - g) \Pi_S^m w \, dx + s_{\partial,\lambda,h} (v - \Pi_{\partial,h}^m v, w - \Pi_{\partial,h}^m w), \\ \langle \mathcal{A}_h^w(v), w \rangle &:= \langle \mathcal{A}_h(v), w \rangle + \eta \kappa^{-1} \langle \mathcal{A}_h^b(v), w \rangle.\end{aligned}$$

We consider the scheme

$$\text{find } u_{h,w} \in V_h \text{ s.t. } \forall v \in V_h, \langle \mathcal{A}_h^w(u_{h,w}), v \rangle = 0. \quad (21)$$

In our analysis of the scheme (21), we rely on the following generalization of the Miranda-Talenti inequality (18) to functions of V_h , that do not necessarily belong to $H_0^1(\Omega)$.

Proposition 6.2 (Miranda-Talenti inequality, boundary penalty case). *For any $v \in V_h$,*

$$|v|_{*,\lambda}^2 - \|\mathcal{L}_\lambda v\|_0^2 \lesssim |v|_{\partial,\lambda,h} |v|_{*,\lambda}. \quad (22)$$

Proof. We distinguish between the cases $\lambda = 0$ and $\lambda > 0$.

Case $\lambda = 0$.

TODO: we can find $v' \in H^2(\Omega)$ such that $v' = v$ on $\partial\Omega$ and $|v'|_2 \lesssim |v|_{\partial,0,h}$ (using the inverse inequality (12) and other arguments).

Let $v_0 := v - v'$, so that $|v|_2 \leq |v_0|_2 + |v'|_2$. By the standard Miranda-Talenti estimate and then the triangle inequality, $|v_0|_2 \leq \|\Delta v_0\|_0 \leq \|\Delta v\|_0 + \|\Delta v'\|_0$, so $|v|_2 \leq \|\Delta v\|_0 + \|\Delta v'\|_0 + |v'|_2$, thus $|v|_2 - \|\Delta v\|_0 \lesssim |v'|_2 \lesssim |v|_{\partial,0,h}$ and $|v|_2^2 - \|\Delta v\|_0^2 = (|v|_2 - \|\Delta v\|_0)(|v|_2 + \|\Delta v\|_0) \lesssim |v|_{\partial,0,h} |v|_2$.

Case $\lambda > 0$. One has

$$\|\mathcal{L}_\lambda v\|_0^2 = \|\Delta v\|_0^2 - 2\lambda \int_\Omega v \Delta v \, dx + \lambda^2 \|v\|_0^2 = \|\Delta v\|_0^2 + 2\lambda |v|_1^2 + \lambda^2 \|v\|_0^2 - 2\lambda \int_{\partial\Omega} v \nabla v \cdot \mathbf{n} \, dx,$$

so that

$$|v|_{*,\lambda}^2 - \|\mathcal{L}_\lambda v\|_0^2 = |v|_2^2 - \|\Delta v\|_0^2 + 2\lambda \int_{\partial\Omega} v \nabla v \cdot \mathbf{n} \, dx.$$

From the analysis of the case $\lambda = 0$, $|v|_2^2 - \|\Delta v\|_0^2 \lesssim |v|_{\partial,0,h} |v|_2 \leq |v|_{\partial,\lambda,h} |v|_{*,\lambda}$. On the other hand,

$$2\lambda \int_{\partial\Omega} v \nabla v \cdot \mathbf{n} \, dx \lesssim \left(\lambda \sum_{S \in \mathcal{S}_{\partial,h}} h_S^{-1} \|v\|_{0,S}^2 \right)^{1/2} \left(\lambda \sum_{S \in \mathcal{S}_{\partial,h}} h_S \|\nabla v\|_{0,S}^2 \right)^{1/2}.$$

For any $S \in \mathcal{S}_{\partial,h}$, by a standard trace inequality and the inverse inequality (11), one has $h_S \|\nabla v\|_{0,S}^2 \lesssim \|\nabla v\|_{0,K}^2$, where K denotes the cell of $\mathcal{K}_h$ whose S is a side. We deduce that $2\lambda \int_{\partial\Omega} v \nabla v \cdot \mathbf{n} \, dx \lesssim |v|_{\partial,\lambda,h} |v|_{*,\lambda}$, which concludes the proof. $\square$

The new Miranda-Talenti inequality (22) allows us to prove a suitable strong monotonicity estimate for the operator $\mathcal{A}_h^w$.

Lemma 6.3 (strong monotonicity, boundary penalty case). *For η large enough in the sense of Definition 5.1, it holds for any $v_1, v_2 \in V_h$ that*

$$\langle \mathcal{A}_h^w(v_1) - \mathcal{A}_h^w(v_2), v_1 - v_2 \rangle \gtrsim \kappa |v_1 - v_2|_{*,\lambda}^2 + \kappa^{-1} |v_1 - v_2|_{\partial,\lambda,h}^2. \quad (23)$$

230 *Proof.* Let $w := v_1 - v_2$. Proceeding as in the proof of Lemma 5.5, one has

$$\langle \mathcal{A}_h(v_1) - \mathcal{A}_h(v_2), w \rangle \geq \|\mathcal{L}_\lambda w\|_0^2 - (1 - \kappa)|w|_{*,\lambda} \|\mathcal{L}_\lambda w\|_0.$$

231 Similarly, it is easily proved that $\langle \mathcal{A}_h^b(v_1) - \mathcal{A}_h^b(v_2), w \rangle \gtrsim |w|_{\partial,\lambda,h}^2$.

232 Let C denote the hidden constant in (22), so that $|w|_{*,\lambda}^2 - \|\mathcal{L}_\lambda w\|_0^2 \leq C|w|_{\partial,\lambda,h}|w|_{*,\lambda}$. We can
233 assume η to be large enough so that $\eta\kappa^{-1}\langle \mathcal{A}_h^b(v_1) - \mathcal{A}_h^b(v_2), w \rangle \geq C^2\kappa^{-1}|w|_{\partial,\lambda,h}^2$. Then

$$\langle \mathcal{A}_h^w(v_1) - \mathcal{A}_h^w(v_2), w \rangle \geq \|\mathcal{L}_\lambda w\|_0^2 - (1 - \kappa)|w|_{*,\lambda} \|\mathcal{L}_\lambda w\|_0 + C^2\kappa^{-1}|w|_{\partial,\lambda,h}^2,$$

234 and, using Young's inequality,

$$\begin{aligned} \langle \mathcal{A}_h^w(v_1) - \mathcal{A}_h^w(v_2), w \rangle &\geq \kappa \|\mathcal{L}_\lambda w\|_0^2 - \frac{1 - \kappa}{2}(|w|_{*,\lambda}^2 - \|\mathcal{L}_\lambda w\|_0^2) + C^2\kappa^{-1}|w|_{\partial,\lambda,h}^2 \\ &\geq \kappa \|\mathcal{L}_\lambda w\|_0^2 - \frac{C}{2}|w|_{\partial,\lambda,h}|w|_{*,\lambda} + C^2\kappa^{-1}|w|_{\partial,\lambda,h}^2. \end{aligned}$$

235 Starting with the inequality $|w|_{*,\lambda}^2 - \|\mathcal{L}_\lambda w\|_0^2 \leq C|w|_{\partial,\lambda,h}|w|_{*,\lambda}$ and applying Young's inequality,
236 one has $|w|_{*,\lambda}^2 - \|\mathcal{L}_\lambda w\|_0^2 \leq (C^2/2)|w|_{\partial,\lambda,h}^2 + (1/2)|w|_{*,\lambda}^2$, so $\|\mathcal{L}_\lambda w\|_0^2 + (C^2/2)|w|_{\partial,\lambda,h}^2 \geq (1/2)|w|_{*,\lambda}^2$
237 and

$$\langle \mathcal{A}_h^w(v_1) - \mathcal{A}_h^w(v_2), w \rangle \geq \frac{1}{2}\kappa|w|_{*,\lambda}^2 - \frac{C}{2}|w|_{\partial,\lambda,h}|w|_{*,\lambda} + \frac{C^2}{2}\kappa^{-1}|w|_{\partial,\lambda,h}^2.$$

238 We conclude using Young's inequality one more time. $\square$

239 As in section 5, we deduce the existence of a unique solution $u_{h,w} \in V_h$ to the scheme (21)
240 as soon as is η large enough, by applying the Browder-Minty theorem. Adapting the proof of
241 Theorem 5.7, we obtain the following error estimate for the solution $u_{h,w}$.

242 **Theorem 6.4** (error estimate, boundary penalty case). *For η large enough in the sense of*
243 *Definition 5.1, and for any $u_I \in V_h$,*

$$|u_I - u_{h,w}|_{*,\lambda} + \kappa^{-1}|u_I - u_{h,w}|_{\partial,\lambda} \lesssim \eta\kappa^{-1}(|u - u_I|_{*,\lambda} + |u - u_I|_{\partial,\lambda,h} + |u - \Pi_h^m u|_{*,\lambda,h}).$$

244 *Proof.* Let $w := u_I - u_{h,w}$. Using the strong monotonicity estimate (23) and then the fact that
245 $u_{h,w}$ is solution to the scheme (21),

$$(|w|_{*,\lambda} + \kappa^{-1}|w|_{\partial,\lambda,h})^2 \lesssim \kappa^{-1}\langle \mathcal{A}_h^w(u_I) - \mathcal{A}_h^w(u_{h,w}), w \rangle = \kappa^{-1}\langle \mathcal{A}_h^w(u_I), w \rangle.$$

246 It remains to prove a suitable upper bound on $\langle \mathcal{A}_h^w(u_I), w \rangle$.

247 One has $\langle \mathcal{A}_h^w(u_I), w \rangle = \langle \mathcal{A}_h^w(u_I) - \mathcal{A}_h^w(\Pi_h^m u), w \rangle + \langle \mathcal{A}_h^w(\Pi_h^m u), w \rangle$. Using the Lipschitz
248 continuity estimate (17), the stabilization scaling inequalities (13) and (20), and the triangle
249 inequality, it is easy to prove that

$$\begin{aligned} \langle \mathcal{A}_h^w(u_I) - \mathcal{A}_h^w(\Pi_h^m u), w \rangle &\lesssim \eta(|u - u_I|_{*,\lambda} + |u - \Pi_h^m u|_{*,\lambda,h})|w|_{*,\lambda} \\ &\quad + \eta\kappa^{-1}(|u - u_I|_{\partial,\lambda,h} + |u - \Pi_h^m u|_{\partial,\lambda,h})|w|_{\partial,\lambda,h}. \end{aligned}$$

250 By standard trace inequalities and then standard polynomial projection estimates, $|u - \Pi_h^m u|_{\partial,\lambda,h} \lesssim$
251 $(h^{-2} + \lambda^{1/2}h^{-1})\|u - \Pi_h^m u\|_{0,h} + (h^{-1} + \lambda^{1/2})|u - \Pi_h^m u|_{1,h} \lesssim |u - \Pi_h^m u|_{*,\lambda,h}$, and thus

$$\langle \mathcal{A}_h^w(u_I) - \mathcal{A}_h^w(\Pi_h^m u), w \rangle \lesssim \eta(|u - u_I|_{*,\lambda} + |u - u_I|_{\partial,\lambda,h} + |u - \Pi_h^m u|_{*,\lambda,h})(|w|_{*,\lambda} + \kappa^{-1}|w|_{\partial,\lambda,h}).$$

252 It remains to prove a suitable upper bound on $\langle \mathcal{A}_h^w(\Pi_h^m u), w \rangle$.

253 Using that $\Pi_h^m u \in \mathbb{P}_{m,h}(\Omega)$, one can compute that

$$\langle \mathcal{A}_h^w(\Pi_h^m u), w \rangle = \int_{\Omega} F_{\gamma}[\Pi_h^m u] \mathcal{L}_{\lambda,h} w \, dx + \eta \kappa^{-1} \sum_{S \in \mathcal{S}_{\partial,h}} (h_S^{-3} + \lambda^2 h_S) \int_S (\Pi_h^m u - g) \Pi_S^m w \, dx.$$

254 Since u is a strong solution to (1) and (2), one has $F_{\gamma}[u] = 0$ a.e. on Ω and $u = g$ a.e. on $\partial\Omega$, so

$$\langle \mathcal{A}_h^w(\Pi_h^m u), w \rangle = \int_{\Omega} (F_{\gamma}[\Pi_h^m u] - F_{\gamma}[u]) \mathcal{L}_{\lambda,h} w \, dx + \eta \kappa^{-1} \sum_{S \in \mathcal{S}_{\partial,h}} (h_S^{-3} + \lambda^2 h_S) \int_S (\Pi_h^m u - u) \Pi_S^m w \, dx.$$

255 Using the Lipschitz continuity estimate (17) for F_{γ} , we deduce that

$$\langle \mathcal{A}_h^w(\Pi_h^m u), w \rangle \lesssim |u - \Pi_h^m u|_{*,\lambda,h} |w|_{*,\lambda} + \eta \kappa^{-1} |u - \Pi_h^m u|_{\partial,\lambda,h} |w|_{\partial,\lambda,h}.$$

256 Using once again that $|u - \Pi_h^m u|_{\partial,\lambda,h} \lesssim |u - \Pi_h^m u|_{*,\lambda,h}$,

$$\langle \mathcal{A}_h^w(\Pi_h^m u), w \rangle \lesssim \eta |u - \Pi_h^m u|_{*,\lambda,h} (|w|_{*,\lambda} + \kappa^{-1} |w|_{\partial,\lambda,h}),$$

257 which concludes the proof. □

258 7 Numerical experiments

259 **TODO.**

260 A (temporary appendix) Different versions of the scheme

261 The aim of this appendix is to give an overview of the different versions of the scheme that we
 262 may want to consider. For now, let us only consider the case of the strong imposition of the
 263 boundary condition.

264 A.1 m, m, m projections

265 This is the version of the scheme currently described in the main part of the draft. One may
 266 choose $s_{*,\lambda,h}$ such that the scheme writes as

$$\begin{aligned} \int_{\Omega} F_{\gamma,h}[u_h] \mathcal{L}_{\lambda,h} v + \eta s_{2,h} (u_h - \Pi_h^m u_h, v - \Pi_h^m v) \\ + \eta \lambda^2 s_{0,h} (u_h - \Pi_h^m u_h, v - \Pi_h^m v) = 0, \quad \forall v \in V_h^0, \end{aligned}$$

267 where, for $v \in V_h$,

$$s_{2,h}(v - \Pi_h^m v, v - \Pi_h^m v) \approx |v - \Pi_h^m v|_{2,h}^2, \quad s_{0,h}(v - \Pi_h^m v, v - \Pi_h^m v) \approx \|v - \Pi_h^m v\|_0^2,$$

268 and, from (14),

$$\begin{aligned} F_{\gamma,h}[v] &= \inf_{\alpha \in \mathcal{A}} \sup_{\beta \in \mathcal{B}} [\gamma^{\alpha,\beta} (\mathcal{L}_h^{\alpha,\beta} v - f^{\alpha,\beta})], \\ \mathcal{L}_h^{\alpha,\beta} v &= A^{\alpha,\beta} : \Pi_h^m \nabla^2 v + b^{\alpha,\beta} \cdot \Pi_h^m \nabla v - c^{\alpha,\beta} \Pi_h^m v, \\ \mathcal{L}_{\lambda,h} v &= \Pi_h^m (\Delta v - \lambda v). \end{aligned}$$

269 **A.2** $m - 2, m - 2, m - 2$ projections

270 We now replace the definitions of the operators $\mathcal{L}_h^{\alpha,\beta}$ and $\mathcal{L}_{\lambda,h}$ as follows:

$$\mathcal{L}_h^{\alpha,\beta} v = A^{\alpha,\beta} : \Pi_h^{m-2} \nabla^2 v + b^{\alpha,\beta} \cdot \Pi_h^{m-2} \nabla v - c^{\alpha,\beta} \Pi_h^{m-2} v, \quad \mathcal{L}_{\lambda,h} v = \Pi_h^{m-2} (\Delta v - \lambda v),$$

271 and we consider the scheme

$$\begin{aligned} \int_{\Omega} F_{\gamma,h}[u_h] \mathcal{L}_{\lambda,h} v + \eta s_{2,h}(u_h - \Pi_h^m u_h, v - \Pi_h^m v) \\ + \eta \lambda^2 s_{0,h}(u_h - \Pi_h^{m-2} u_h, v - \Pi_h^{m-2} v) = 0, \quad \forall v \in V_h^0. \end{aligned}$$

272 Observe that the projection operators in the $s_{2,h}$ stabilization term are still to the degree m , and
273 not $m - 2$.

274 **Strong monotonicity analysis.** Proceeding as in the proof of Lemma 5.5, for $v_1, v_2 \in V_h^g$
275 and $w = v_1 - v_2$,

$$\langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle \geq \|\Pi_h^{m-2} \mathcal{L}_{\lambda} w\|_0^2 - (1 - \kappa) |w|_{*,\lambda} \|\mathcal{L}_{\lambda} w\|_0.$$

276 On the other hand,

$$\begin{aligned} \langle \mathcal{A}_h^s(v_1) - \mathcal{A}_h^s(v_2), w \rangle &\gtrsim |w - \Pi_h^m w|_{2,h}^2 + \lambda^2 \|w - \Pi_h^{m-2} w\|_0^2 \\ &\gtrsim \|\Delta w - \Delta \Pi_h^m w\|_0^2 + \lambda^2 \|w - \Pi_h^{m-2} w\|_0^2 \\ &\geq \|\Delta w - \Pi_h^{m-2} \Delta w\|_0^2 + \lambda^2 \|w - \Pi_h^{m-2} w\|_0^2 \\ &\geq \|\Delta w - \Pi_h^{m-2} \Delta w - \lambda(w - \Pi_h^{m-2} w)\|_0^2 \\ &= \|\mathcal{L}_{\lambda} w - \Pi_h^{m-2} \mathcal{L}_{\lambda} w\|_0^2, \end{aligned}$$

277 so one can conclude as in the proof of Lemma 5.5.

278 **A.3** $m - 2, m - 1, m$ projections

279 Finally, we replace the definitions of the operators $\mathcal{L}_h^{\alpha,\beta}$ and $\mathcal{L}_{\lambda,h}$ as follows:

$$\mathcal{L}_h^{\alpha,\beta} v = A^{\alpha,\beta} : \Pi_h^{m-2} \nabla^2 v + b^{\alpha,\beta} \cdot \Pi_h^{m-1} \nabla v - c^{\alpha,\beta} \Pi_h^m v, \quad \mathcal{L}_{\lambda,h} v = \Pi_h^{m-2} \Delta v - \lambda \Pi_h^m v,$$

280 and we consider the scheme

$$\begin{aligned} \int_{\Omega} F_{\gamma,h}[u_h] \mathcal{L}_{\lambda,h} v + \eta \kappa^{-1} s_{2,h}(u_h - \Pi_h^m u_h, v - \Pi_h^m v) \\ + \eta \lambda^2 s_{0,h}(u_h - \Pi_h^m u_h, v - \Pi_h^m v) = 0, \quad \forall v \in V_h^0. \end{aligned}$$

281 Notice the factor κ^{-1} appearing in front of the $s_{2,h}$ stabilization term.

282 **Strong monotonicity analysis.** Proceeding as in the proof of Lemma 5.5, for $v_1, v_2 \in V_h^g$
283 and $w = v_1 - v_2$,

$$\langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle \geq \|\mathcal{L}_{\lambda,h} w\|_0^2 - (1 - \kappa) |w|_{*,\lambda} \|\mathcal{L}_{\lambda,h} w\|_0.$$

284 Using the Miranda-Talenti estimate (18),

$$\langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle \geq \|\mathcal{L}_{\lambda,h} w\|_0^2 - (1 - \kappa) \|\mathcal{L}_{\lambda} w\|_0 \|\mathcal{L}_{\lambda,h} w\|_0.$$

Using that $\mathcal{L}_{\lambda,h}w = \Pi_h^m \mathcal{L}_\lambda w - (\Pi_h^m \Delta w - \Pi_h^{m-2} \Delta w)$ and

$$\|\Pi_h^m \mathcal{L}_\lambda w\|_0 \leq \|\mathcal{L}_\lambda w\|_0, \quad \|\Pi_h^m \Delta w - \Pi_h^{m-2} \Delta w\|_0 \leq \|\Delta w - \Pi_h^{m-2} \Delta w\|_0,$$

one has

$$\begin{aligned} \|\mathcal{L}_{\lambda,h}w\|_0^2 &= \|\Pi_h^m \mathcal{L}_\lambda w\|_0^2 + \|\Pi_h^m \Delta w - \Pi_h^{m-2} \Delta w\|_0^2 - 2 \int_{\Omega} \Pi_h^m \mathcal{L}_\lambda w (\Pi_h^m \Delta w - \Pi_h^{m-2} \Delta w) dx \\ &\geq \|\Pi_h^m \mathcal{L}_\lambda w\|_0^2 - 2 \int_{\Omega} \Pi_h^m \mathcal{L}_\lambda w (\Pi_h^m \Delta w - \Pi_h^{m-2} \Delta w) dx \\ &\geq \left(1 - \frac{\kappa}{3}\right) \|\Pi_h^m \mathcal{L}_\lambda w\|_0^2 - \frac{3}{\kappa} \|\Delta w - \Pi_h^{m-2} \Delta w\|_0^2 \end{aligned}$$

and

$$\begin{aligned} (1 - \kappa) \|\mathcal{L}_\lambda w\|_0 \|\mathcal{L}_{\lambda,h}w\|_0 &\leq (1 - \kappa) \|\mathcal{L}_\lambda w\|_0^2 + (1 - \kappa) \|\mathcal{L}_\lambda w\|_0 \|\Delta w - \Pi_h^{m-2} \Delta w\|_0 \\ &\leq \left(1 - \kappa + \frac{\kappa}{3}\right) \|\mathcal{L}_\lambda w\|_0^2 + \frac{3(1 - \kappa)^2}{4\kappa} \|\Delta w - \Pi_h^{m-2} \Delta w\|_0^2 \\ &\leq \left(1 - \frac{2\kappa}{3}\right) \|\mathcal{L}_\lambda w\|_0^2 + \frac{3}{4\kappa} \|\Delta w - \Pi_h^{m-2} \Delta w\|_0^2, \end{aligned}$$

hence

$$\langle \mathcal{A}_h^c(v_1) - \mathcal{A}_h^c(v_2), w \rangle \geq \left(1 - \frac{\kappa}{3}\right) \|\Pi_h^m \mathcal{L}_\lambda w\|_0^2 - \left(1 - \frac{2\kappa}{3}\right) \|\mathcal{L}_\lambda w\|_0^2 - \frac{15}{4\kappa} \|\Delta w - \Pi_h^{m-2} \Delta w\|_0^2.$$

One may conclude by proceeding similarly to the proof of Lemma 5.5.

B (temporary appendix) Some nonlinear problems

Let us consider an equation of the form

$$\inf_{\alpha \in \mathcal{A}} [A^\alpha(x) : \nabla^2 u(x) + b^\alpha(x) \cdot \nabla u(x) - c^\alpha(x)u(x) - f^\alpha(x)] = 0 \quad \text{a.e. in } \Omega.$$

In order to solve the associated numerical scheme, one has to be able, for given $M \in \mathcal{S}_d^{++}$, $p \in \mathbb{R}^d$, $r \in \mathbb{R}$, and $x \in \mathbb{R}^d$, to solve the minimization problem

$$\min_{\alpha \in \mathcal{A}} [A^\alpha(x) : M + b^\alpha(x) \cdot p - c^\alpha(x)r - f^\alpha(x)].$$

In this appendix, we study the above minimization problem in some specific cases.

B.1 Second experiment in [5]

One has $\mathcal{A} = \text{SO}(2)$, $A^\alpha(x) = \alpha^\top A_0 \alpha$ for some $A_0 \in \mathcal{S}_d$, and $b^\alpha(x)$, $c^\alpha(x)$, and $f^\alpha(x)$ do not depend on α . Then the minimization problem reduces to

$$\min_{\alpha \in \text{SO}(2)} (\alpha^\top A_0 \alpha) : M.$$

I believe it is easily verified that the minimal value is $\lambda_{\min}(A_0)\lambda_{\min}(M) + \lambda_{\max}(A_0)\lambda_{\max}(M)$, and that, if $v_{\min}$ and $v_{\max}$ denote unit eigenvectors of M associated respectively to the eigenvalues $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$, then the optimal α is such that

$$\alpha^\top A_0 \alpha = \lambda_{\min}(A_0)v_{\min}v_{\min}^\top + \lambda_{\max}(A_0)v_{\max}v_{\max}^\top.$$

B.2 First experiment in [5]

One has $\mathcal{A} = [0, \pi/3] \times \text{SO}(2)$,

$$A^{(\theta, R)}(x) = \frac{1}{2} R^\top A_0(\theta) R, \quad \text{where } A_0(\theta) := \begin{pmatrix} 1 + \sin^2(\theta) & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \cos^2 \theta \end{pmatrix},$$

$f^{(\theta, R)}(x) = \sqrt{3} \sin^2 \theta / \pi^2 + g(x)$, and $b^{(\theta, R)}$ and $c^{(\theta, R)}$ do not depend on θ and R . Then the minimization problem reduces to

$$\min_{\theta \in [0, \pi/3]} \min_{R \in \text{SO}(2)} \left(\frac{1}{2} (R^\top A_0(\theta) R) : M - \frac{\sqrt{3}}{\pi^2} \sin^2 \theta \right).$$

One may proceed as above for the minimization with respect to R . One has

$$\det(A_0(\theta) - \lambda I_2) = (1 + \sin^2 \theta - \lambda)(\cos^2 \theta - \lambda) - \sin^2 \theta \cos^2 \theta = \lambda^2 - 2\lambda + \cos^2 \theta,$$

so that $\lambda_{\min}(A_0(\theta)) = 1 - \sin \theta$ and $\lambda_{\max}(A_0(\theta)) = 1 + \sin \theta$. Thus, the minimization problem reduces to

$$\min_{\theta \in [0, \pi/3]} \left(\frac{1}{2} (1 - \sin \theta) \lambda_{\min}(M) + \frac{1}{2} (1 + \sin \theta) \lambda_{\max}(M) - \frac{\sqrt{3}}{\pi^2} \sin^2 \theta \right),$$

where the minimum in R was reached by an R for which

$$R^\top A_0(\theta) R = (1 - \sin \theta) v_{\min} v_{\min}^\top + (1 + \sin \theta) v_{\max} v_{\max}^\top,$$

where $v_{\min}$ and $v_{\max}$ denote unit eigenvectors of M associated respectively to the eigenvalues $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$. The one-dimensional minimization problem with respect to θ can then be solved by standard techniques.

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